INDIAN PATENT OFFICE

FORM 2 — COMPLETE SPECIFICATION

PATENT APPLICATION — Form 2: Specification

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> ***NOTICE*** *This document is a drafting template for a complete specification filed at the Indian Patent Office. It is structured per the Patents Act, 1970, the Patents Rules, 2003, and the Form 2 (Complete Specification) layout.* ***It is not legal advice.*** *Before filing, engage a registered patent agent (a list is at <https://ipindia.gov.in>) to: 1. Conduct a prior-art search against existing patents (especially KR/JP/CN filings on AI yoga from 2018–2024). 2. Prepare and notarise drawings (Figures 1–4 are referenced but must be drafted as line drawings on A4 sheets). 3. File via the e-filing portal (<https://ipindiaonline.gov.in>) along with Form 1 (Application), Form 5 (Declaration of Inventorship), and government fee (₹1,600 for an individual / Startup vs ₹8,000 for a company). 4. Decide whether to file a Provisional (12 months grace, ~70% cheaper) before this Complete Specification.*

FORM 2 — THE PATENTS ACT, 1970 — Complete Specification

TITLE OF THE INVENTION

A BROWSER-NATIVE PRIVACY-PRESERVING SYSTEM AND METHOD FOR HYSTERESIS-FILTERED REAL-TIME YOGA POSE COACHING WITH PRIORITY-RESOLVED VOICE COMMAND GRAMMAR

(a) FIELD OF THE INVENTION

The present invention relates to the field of computer-assisted physical instruction, more particularly to systems and methods for real-time pose-correction feedback during yoga practice using on-device machine learning inference and voice-driven hands-free interaction. The invention further relates to multi-tenant architecture for deploying such systems at institutional scale.

(b) BACKGROUND OF THE INVENTION

Existing AI-assisted yoga systems suffer from at least the following deficiencies:

(i) **Native-app distribution dependence.** Conventional yoga apps require installation through Apple App Store, Google Play Store, or proprietary device sideloading, creating administrative overhead for institutional rollout (educational institutions, corporate wellness programmes) and excluding low-end devices that lack the storage to host such applications.

(ii) **Cloud-side pose inference.** Existing systems that perform pose estimation on a remote server transmit raw video frames or extracted skeletal data over the network. This (a) introduces latency that defeats real-time coaching, (b) requires reliable broadband connectivity inappropriate for users in low-connectivity regions, and (c) creates a privacy liability — biometric data is regulated under the Digital Personal Data Protection Act, 2023.

(iii) **Frame-level feedback chatter.** Conventional rule-based pose scoring emits an audio correction whenever a rule fails on the current frame and rescinds it whenever the rule passes, producing oscillating spoken feedback (e.g., "lift your knee" and "knee corrected" repeating several times per second) that is cognitively burdensome and counter-productive to practice.

(iv) **Voice-command ambiguity.** Existing voice-controlled fitness applications cannot distinguish between *action commands* (e.g., "next pose", "what is my score?") and *content references* (e.g., "warrior two" — referring to the pose Virabhadrasana II). When the spoken token "two" is parsed as a numeral, the action grammar fails, and when the action grammar takes priority, the user cannot verbally select certain poses.

(v) **Lack of institutional partition.** Existing systems are designed for individual consumers; they do not support a tenant-isolated data model in which a college can onboard 1,000 students with strict guarantees that no other tenant's data is accessible.

There exists, therefore, a need for a yoga coaching system that (a) runs natively in a browser without app installation, (b) executes pose inference on-device for privacy and latency, (c) suppresses frame-level feedback chatter through stateful filtering, (d) disambiguates voice commands deterministically, and (e) supports multi-tenant institutional deployment.

The present invention addresses each of the above deficiencies.

(c) OBJECTS OF THE INVENTION

The principal objects of the present invention are:

1 To provide a yoga coaching system that operates in any modern web browser without requiring native application installation.

1 To provide such a system in which all biometric inference (pose estimation, alignment scoring) is performed on the user's local computing device and no raw video, image, or skeletal keypoint data is transmitted to any remote server.

1 To provide a method of converting per-frame pose rule evaluations into spoken corrections through a per-rule fail/pass debounce filter (hereinafter "hysteresis filter") that suppresses transient detection artefacts and emits feedback only after a sustained condition is observed.

1 To provide a voice-command interpretation method that resolves ambiguity between action commands and content references through priority-ordered regular expression matching with a fallback fuzzy pose-name lookup.

1 To provide a multi-tenant institutional deployment model in which data is partitioned by organisational email domain with route-level access enforcement.

1 To provide a six-semester certificate programme integrating real-time pose evaluation with quiz-based knowledge assessment, issuing cryptographically-signed completion certificates.

(d) SUMMARY OF THE INVENTION

The present invention discloses a system and method for real-time yoga pose coaching, comprising:

(i) a *client device* (a personal computer, laptop, tablet, or mobile phone) executing a web browser supporting the WebGL and WebSpeech APIs;

(ii) a *pose estimation module* loaded via the said web browser, comprising a convolutional neural network for single-person pose estimation (in the preferred embodiment, MoveNet SinglePose Thunder) executing within a JavaScript runtime environment using WebGL hardware acceleration;

(iii) a *rule engine* comprising a finite, per-pose set of predicates over the smoothed keypoints, where each predicate is either an angle constraint between three keypoints or a distance- ratio constraint between two pairs of keypoints, and is associated with a natural-language correction string;

(iv) a *hysteresis filter* comprising, for each predicate, a fail counter and a pass counter, and a state flag taking values in the set {silent, speaking}, wherein: (a) a correction is emitted to a speech synthesis subsystem only when the fail counter for the corresponding rule equals or exceeds a fail threshold (default seven frames); (b) the state flag transitions from speaking back to silent only when the pass

counter equals or exceeds a pass threshold (default five frames);

(v) a *voice command grammar* comprising a priority-ordered list of regular expressions, each mapped to a fixed action handler, with a fallback fuzzy lookup against a catalogue of pose names;

(vi) a *server-side multi-tenant data layer* in which records are partitioned by an organisational email domain identifier, and read/write operations are gated by a request-level access decorator that injects the requesting user's organisational identifier into the database query;

(vii) a *certification module* that records per-pose alignment sessions and quiz outcomes, computes per-semester completion, and issues cryptographically signed PDF certificates.

The said hysteresis filter is the technical heart of the invention; the priority-resolved voice grammar is the second core contribution; the multi-tenant model is the third.

(e) BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings (to be prepared as A4 line drawings) illustrate:

- **Figure 1.** System architecture: client device ↔ web server ↔ multi-tenant database, with the on-device pose-estimation pipeline highlighted as residing entirely client-side.
- **Figure 2.** Hysteresis filter state diagram: states {silent, speaking} with transitions labelled (fail≥F → speak), (pass≥P → silent).
- **Figure 3.** Priority-ordered voice command resolution flow: utterance → action regex matching → fallback pose lookup → action dispatch.
- **Figure 4.** Multi-tenant data isolation: per-request access decorator injecting `organizationEmail` into all collection queries.

(f) DETAILED DESCRIPTION OF THE INVENTION

1. The Client-Side Pose Pipeline

[Detailed description of the per-frame pipeline as in Section III.B of the research paper. Refer to Figure 1.]

2. The Rule Grammar (Claim 1 / Claim 6 basis)

A pose `P` is characterised by:

```
P = (N, S, R, T)
    N — Sanskrit name
    S — stance class ∈ {stand, seat, prone, supine}
    R — finite rule set {r_1, ..., r_k}
    T — target hold duration (seconds)

r_i = (ϕ_i, τ_i, c_i)
    ϕ_i — predicate over smoothed keypoints
    τ_i — confidence threshold
    c_i — natural-language correction string

ϕ ::= angle(K_a, K_b, K_c) ∈ [θ_min, θ_max]
    | dist(K_a, K_b) / dist(K_c, K_d) ∈ [ρ_min, ρ_max]
```

Twenty curated poses are encoded in this grammar; the complete rule sets are in Annexure A.

3. The Hysteresis Filter (Independent Claim 1)

The state machine for each rule i:

```
state[i]      ∈ {silent, speaking}    initially silent
fail_cnt[i]   ∈ ■                     initially 0
pass_cnt[i]   ∈ ■                     initially 0
F             = 7   (fail threshold)
P             = 5   (pass threshold)
```

On each frame at time t:

```
if NOT ϕ_i(p■_t):                      // rule failing
    fail_cnt[i] := fail_cnt[i] + 1
    pass_cnt[i] := 0
    if state[i] = silent and fail_cnt[i] ≥ F:
        emit_speech(c_i)
        state[i] := speaking
else:                                  // rule passing
    pass_cnt[i] := pass_cnt[i] + 1
    fail_cnt[i] := 0
    if state[i] = speaking and pass_cnt[i] ≥ P:
        state[i] := silent
```

The asymmetry F > P is intentional: the system enters the speaking state cautiously (requires 280 ms of failure at 25 fps) but exits it quickly (requires 200 ms of passing) — rewarding the user's correction with rapid silence.

4. The Voice Command Grammar (Independent Claim 2)

```
RESOLVE(utterance):
    for each (regex_j, handler_j) in PRIORITY_TABLE:
        if regex_j.test(utterance):
            return handler_j(utterance)
    pose_name := fuzzy_match(utterance, ASANA_LIBRARY)
    if pose_name is not null:
        return select_pose(pose_name)
    return error_unrecognised
```

Where PRIORITY_TABLE is a static ordered list of 12 (regex, handler) pairs covering: end session, score query, next/previous, reference overlay, repeat, voice toggle, breath toggle, diagnostics, hold timer, and pose modifications. The fallback fuzzy match uses a normalised Levenshtein distance against the canonical Sanskrit and English aliases of the 20 poses in the library.

5. The Multi-Tenant Data Layer (Independent Claim 3)

Each request to a protected route passes through a decorator that:

```
def require_org(handler):
    def wrapper(request):
        user = authenticate(request)
        if user.role not in {"org_admin", "student"}: deny()
        request._org_email := user.organizationEmail
        return handler(request)
    return wrapper
```

All MongoDB queries within `handler` then filter by `{ organizationEmail: request._org_email }`. There is no code path in the application by which an org_admin can query data outside their `organizationEmail` scope.

6. The Certification Module (Independent Claim 4)

A pose `P` is *passed* by user `u` when:

```
∃ session s:
    s.user = u, s.pose = P, max_duration_at(score ≥ 90%, s) ≥ T_P

AND

∃ quiz_attempt q:
    q.user = u, q.pose = P, q.score = 100%
```

A semester is *passed* when all its poses are passed. The master certificate is *issued* when all six semesters are passed; the certificate is a PDF signed with an RSA-2048 key whose public half is published at a verification URL embedded in the PDF.

(g) CLAIMS

We claim:

1. A method for real-time yoga pose coaching, comprising:

(a) capturing a video stream from a user-facing camera of a client device;

(b) executing a single-person convolutional pose-estimation neural network within a web browser running on the said client device, said neural network producing a set of keypoint coordinates and confidence scores per frame;

(c) smoothing said keypoint coordinates by an exponential moving average to obtain smoothed keypoints;

(d) evaluating, against the said smoothed keypoints, a finite set of pose-specific rule predicates each comprising either an angle constraint over three keypoints or a distance-ratio constraint over two pairs of keypoints;

(e) maintaining, for each said rule predicate, a fail counter and a pass counter, and a state flag taking values in {silent, speaking};

(f) emitting a corresponding pre-defined natural-language correction string to a speech synthesis subsystem if and only if the said state flag for the said rule predicate is silent AND the said fail counter exceeds a fail threshold;

(g) transitioning the said state flag from speaking back to silent if and only if the said pass counter exceeds a pass threshold;

characterised in that the said fail threshold is strictly greater than the said pass threshold, and the said keypoint inference of step (b) is performed entirely on the said client device with no raw video, image, or keypoint data transmitted to any remote server.

2. A method for resolving voice commands during yoga practice, comprising:

(a) receiving a recognised speech utterance from a speech- recognition subsystem of a web browser;

(b) testing the said utterance against a priority-ordered list of regular expressions each mapped to a respective action handler;

(c) invoking the action handler corresponding to the first said regular expression that matches the said utterance;

(d) and only if no said regular expression matches, performing a fuzzy lexical match of the said utterance against a catalogue of pose names and invoking a pose-selection action;

characterised in that the said priority-ordered list disambiguates utterances containing tokens that are simultaneously valid as action keywords and as pose-name components by deterministically resolving in favour of the action.

3. A multi-tenant deployment system for the method of claim 1, comprising:

(a) a server-side data store partitioned by an organisational email-domain identifier;

(b) an authentication subsystem that associates each authenticated user with exactly one said organisational identifier;

(c) an access-control decorator applied to each protected route handler that injects the said organisational identifier into database query filters,

characterised in that no application code path permits a request authenticated under one said organisational identifier to read or modify data associated with a different said organisational identifier.

4. A certification method for use with the method of claim 1, comprising:

(a) recording, for each pose-practice session, the maximum duration during which the said alignment score (computed as the fraction of said rule predicates passing) remained at or above ninety percent;

(b) recording, for each said pose, the score on a fixed-question anatomical quiz;

(c) marking the said pose as passed when the said maximum duration at or above ninety percent is at least ten seconds AND the said quiz score is one hundred percent;

(d) marking a semester as passed when all poses assigned to the said semester are marked passed;

(e) issuing a cryptographically signed PDF certificate when all six semesters of a fixed curriculum are marked passed.

5. The method of claim 1 wherein the said exponential moving average smoothing parameter is in the range [0.5, 0.7] and the said pose-estimation neural network is MoveNet SinglePose Thunder.

6. The method of claim 1 wherein the said fail threshold is seven frames and the said pass threshold is five frames at a nominal frame rate of twenty-five frames per second.

7. The method of claim 1 wherein the said rule predicates for asymmetric poses are parameterised by a leading-side flag and the said voice command grammar of claim 2 accepts said leading-side as an optional suffix to the said pose name.

8. The system of claim 3 further comprising a bulk-onboarding subsystem that ingests an Excel spreadsheet of student records, validates each record against the said organisational domain, generates a single-use invitation token per record, and dispatches a magic-link email containing the said token.

9. A non-transitory computer-readable medium storing instructions which, when executed by one or more processors of a client device having a camera and a microphone, cause the said processors to perform the method of claim 1.

10. A system comprising one or more client devices and a server, configured to perform the method of claim 1 in combination with the multi-tenant deployment system of claim 3 and the certification method of claim 4.

(h) ABSTRACT OF THE DISCLOSURE

Disclosed is a browser-native system and method for real-time yoga pose coaching. A web-based client device captures camera video, executes an on-device convolutional pose-estimation neural network, and evaluates the detected keypoints against per-pose rule predicates. A novel hysteresis filter — a per-rule fail/pass debounce state machine with asymmetric thresholds (fail strictly greater than pass) — suppresses frame-level feedback chatter and emits spoken corrections only after a sustained failure is observed, returning to silence only after a sustained recovery. A priority-ordered voice command grammar deterministically disambiguates action commands from content references. A server-side multi-tenant data layer partitions student records by organisational email domain with route-level access enforcement. A certification module integrates pose practice and quiz performance to issue cryptographically signed completion certificates for a fixed six-semester twenty-pose curriculum. No raw video, image, or keypoint data is transmitted from the client device.

ANNEXURES (to be appended)

- **Annexure A.** Complete rule sets for the twenty curated asanas (formatted as a table; rows include pose name, stance class, rule predicate, threshold range, correction string).
- **Annexure B.** Example regex priority table for the voice command grammar.
- **Annexure C.** Example bulk-onboarding Excel template format.
- **Annexure D.** Sample certificate PDF showing the embedded verification URL and the RSA signature block.

This specification is filed in respect of an invention; the applicant requests that a patent may be granted to him for the invention described and claimed herein.

Date. [DD/MM/2026]

Signature of applicant. _____

Name. P. G. Ayush Rai

_____

Signature of applicant P. G. Ayush Rai

Dated this _____ day of _____, 2026, at Mysuru, India.